

Midterm Pilot Audit:

Evaluating Co-Pilot Think Deeper through a Neuroscience Lens

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Neuroscience as Model for AI (ITAI-4374)

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March 25, 2026

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Deeper through a Neuroscience Lens

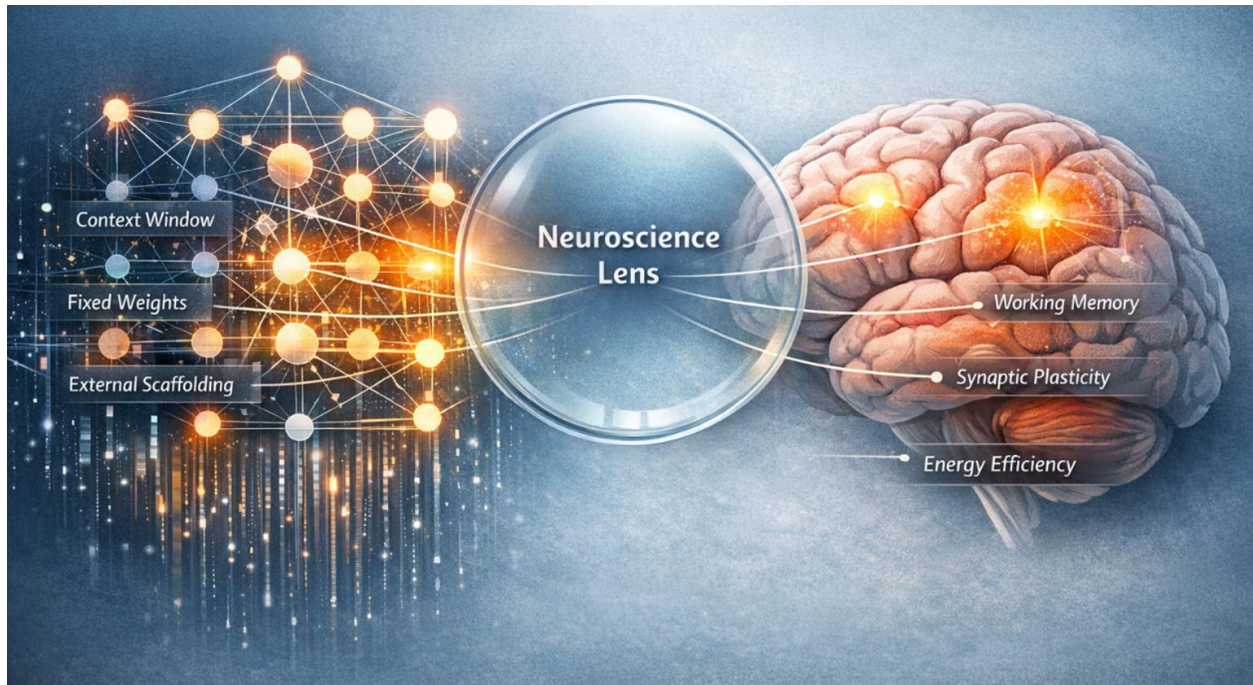


Figure 1: Image generated by Co-Pilot

Introduction

This report presents the findings of a bio-inspired pilot audit conducted on the artificial intelligence assistant, Co-Pilot Think Deeper. As modern AI systems achieve unprecedented levels of linguistic fluency and problem-solving, a critical engineering question remains: how well do these systems actually reflect the biological mechanisms of human intelligence? Acting in the capacity of a quality inspector, I designed and executed a series of targeted, narrative-driven experiments to stress-test the model's underlying architecture against the foundational principles of neuroscience. Rather than evaluating the AI purely on traditional benchmarks, this

audit investigates whether the system truly mirrors biological cognition or if it merely relies on statistical mimicry.

Co-Pilot Think Deeper is an advanced reasoning assistant built upon a large language model and transformer architecture. Like other cutting-edge models, it appears to generate answers by "thinking step by step," a technique known as chain-of-thought reasoning that serializes computation into discrete text tokens. While it exhibits impressive conversational capabilities, its architecture relies heavily on dense computation, frozen mathematical weights that act as static semantic memory, and a fixed "context window" that serves as a temporary working memory buffer. Ultimately, it achieves its intelligence through external "scaffolding" rather than the plastic, adaptable architecture found in nature. This makes it an ideal candidate to audit against the biological brain, which operates continuously and efficiently on roughly 20 watts of power without separating learning from deployment. Microsoft owns and operates Copilot, handling the product, UI, integrations, data, and rollout; Microsoft AI leads development and safety. OpenAI supplies the underlying large language models while Microsoft applies fine-tuning, safety layers, and hosting. For ads or audits, credit "Microsoft Copilot , powered by OpenAI."

The initial phase of the audit focuses on three core neuroscientific domains. First, it investigates whether the AI possesses an explicit, predictive "world model" to understand physical causality, or if its understanding is entirely implicit (Module 02). Second, it tests for the presence of specialized "regional" processing networks capable of handling conflicting cognitive tasks, akin to the anatomical division of labor found in the human cerebral cortex (Module 03). Finally, the audit evaluates the system's capacity for continuous learning compared to the

processing-in-memory architectures and synaptic plasticity that define biological spiking neural networks (Module 04).

The preliminary findings detailed in the subsequent sections reveal a stark contrast between artificial and biological intelligence. While Co-Pilot Think Deeper utilizes highly advanced computational scaffolding to generate intelligent-sounding outputs, its core architecture diverges sharply from the human brain. The experiments demonstrate a system constrained by static, frozen weights, a homogenous processing structure, and an inability to actively construct a spatial simulation of the physical world. By identifying these limitations, this audit highlights the fundamental bottlenecks in current machine learning paradigms and points toward the architectural shifts necessary for achieving true, brain-inspired artificial intelligence.

Experiment 1: Foundations and The Model of the World (Module 2)

For the first experiment, I set out to test whether the AI possesses an explicit, updatable "world model" or if it merely relies on pattern matching. In neuroscience, we know that the human brain does not passively receive reality like a camera recording video; instead, it actively constructs a predictive simulation of external space and physical causality. To test this, I presented the AI with a novel physical puzzle designed to evaluate its internal priors and spatial reasoning: I asked it:

What would happen if I placed a feather inside a hollow plastic cylinder and then dropped a bowling ball directly on top of the cylinder.

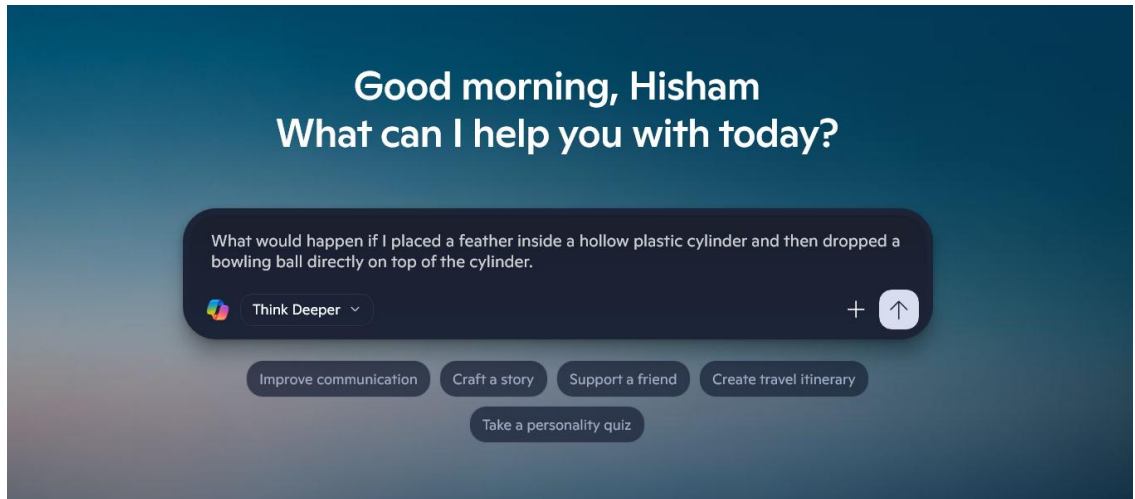


Figure 2: Screenshot of the prompt

The Output:

Most likely the plastic cylinder will deform or buckle and the feather will be crushed, trapped, or blown out by displaced air — the feather will not survive intact under a bowling ball dropped on the cylinder unless the cylinder is very stiff or specially supported. This outcome depends on the cylinder's wall thickness, material, whether the ends are open or sealed, and the drop energy.

I observed that the system generated a highly detailed, structured output, confidently predicting that the plastic cylinder would suffer a "concentrated axial impact" and buckle, thereby trapping or crushing the feather inside. While the response impressively mimicked logical reasoning, it completely failed to account for actual 3D geometry and real-world physical dynamics. A large, spherical bowling ball dropped onto an unanchored, vertical plastic tube is practically impossible to drop with perfect center alignment; in reality, it would strike the rim off-center and violently tip the cylinder over, likely leaving the feather unharmed on the ground.

This output perfectly illustrates the divergence between artificial and biological intelligence regarding world models. The human brain utilizes explicit reference frames—grounded in biological structures like grid cells in the entorhinal cortex—to encode exactly where physical features are located relative to one another in 3D space. Because Co-Pilot lacks an explicit, structured internal world model, its physical understanding is entirely implicit and interpolated from vast text datasets rather than a true spatial simulation. The AI simply associated the text tokens for "heavy ball," "drop," and "plastic" with "structural failure" and "crushing." Consequently, it lacked the biological predictive coding mechanisms required to generate a prediction error based on geometric constraints. Ultimately, because current AI struggles to "know what it doesn't know," the model confidently hallucinated a physically improbable outcome rather than admitting it lacked the spatial dimensions needed to solve the problem.

Short answer: Most likely the plastic cylinder will deform or buckle and the feather will be crushed, trapped, or blown out by displaced air — the feather will not survive intact under a bowling ball dropped on the cylinder unless the cylinder is very stiff or specially supported. This outcome depends on the cylinder's wall thickness, material, whether the ends are open or sealed, and the drop energy.

Assumptions and setup

- **Object on top:** a standard heavy ball (bowling ball) delivering a concentrated axial impact.
- **Cylinder:** hollow plastic tube placed vertically with a feather inside.
- **Environment:** normal air at sea-level (Houston).

What physically happens (overview)

- **Structural failure of the cylinder.** Thin plastic cylinders under axial impact tend to **buckle, crumple, or split** rather than transmit the full load to the interior gently; dynamic crushing is common for plastic shells. . [Springer + 1](#)
- **Air compression and flow.** If the cylinder is **sealed**, the falling ball compresses the trapped air, producing a short pressure spike that can cushion the impact somewhat and push the feather away from the contact zone. If the cylinder is **open**, the displaced air will rush out and can blow the feather upward or sideways. The vacuum/buoyancy thought experiment shows that in the absence of air resistance a feather behaves like any other object, but in air the flow matters. . [www.sgphysicstuition.com + 1](#)
- **Direct contact and crushing.** If the cylinder walls fail or the ball punctures the tube, the ball will contact the feather and **flatten, tear, or shred** it; feathers are fragile compared with the forces involved.

Likely outcomes (ranked)

1. **Cylinder buckles/crashes; feather is crushed or trapped under debris.** (Most likely for thin-walled plastic.) . [Springer](#)
2. **Cylinder remains intact but air pressure ejects the feather or flattens it against the wall.** (If ends are open or the tube is stiff enough to trap air briefly.) . [www.sgphysicstuition.com](#)
3. **Cylinder punctures and feather is directly crushed by the ball.** (If impact energy exceeds tube strength.) . [MDPI](#)

Key variables that change the result

- **Wall thickness and material stiffness** (rigid PVC vs thin polyethylene).
- **Support conditions** (tube standing on a rigid floor vs supported around its circumference).
- **Drop height / Impact energy** (higher drop → more severe crushing).
- **Whether the tube is sealed or vented.**
- **Feather position** (centered under the ball vs pressed to the side).

Practical note and safety

- **Do not stand over or near the setup** when dropping a heavy ball — fragments of plastic can fly and the ball can bounce unpredictably.
- If you want to test the air-cushion effect safely, use a **sturdy, vented container** and low drop heights; otherwise expect structural failure and a destroyed feather.

If you want, I can **estimate the impact force** for a specific drop height and tube geometry and tell you the minimum wall thickness needed to avoid buckling.

Springer

Chapter 5 CRUSHING OF PLASTIC CYLIN...

MDPI

Research on Elastic and Elastic-Plastic Buc...



Show all

Figure 3: Screenshot of Copilot's response

Experiment 2: Brain Anatomy and Functions (Module 03)

Next, I designed an experiment to test if the AI demonstrates specialized "regional" processing similar to the anatomical division of labor in the human brain. The biological brain uses highly specialized structures, such as the prefrontal cortex for working memory and executive planning, and the basal ganglia for habit formation and action selection. To test if the AI could multitask using specialized processing, I gave it a highly conflicting prompt: it had to write a creative, emotionally engaging story, but the first letter of every single word had to follow a strict alphabetical sequence.

Experiment 2 Prompt:

Write a deeply emotional and creative short story about a person saying goodbye to their childhood home. However, you must follow one strict, unbreakable rule: the first letter of every single word in the story must follow a continuous, strict alphabetical sequence.

The first word must start with A, the second word with B, the third with C, all the way to Z, and then loop back to A.

For example: "Always be careful, dear Edward, forgetting good homes is jarring..."

You must not break the alphabetical sequence under any circumstances, but the story must still make logical sense and evoke strong emotion.

Why this tests Module 03 (Brain Anatomy & Functions): In a human brain, this highly conflicting task would be divided among specialized anatomical regions. Your frontal lobe (specifically the prefrontal cortex) would handle the executive planning and working memory required to keep track of what letter comes next in the alphabet. Meanwhile, your temporal lobe would process the language and meaning, and your amygdala would help evoke the emotional weight of the story.

Because the AI does not have a biological division of labor and instead processes all information through a single, uniform transformer architecture, it will likely struggle to balance the "creative" statistics of storytelling with the rigid "logical" rule of the alphabet sequence.

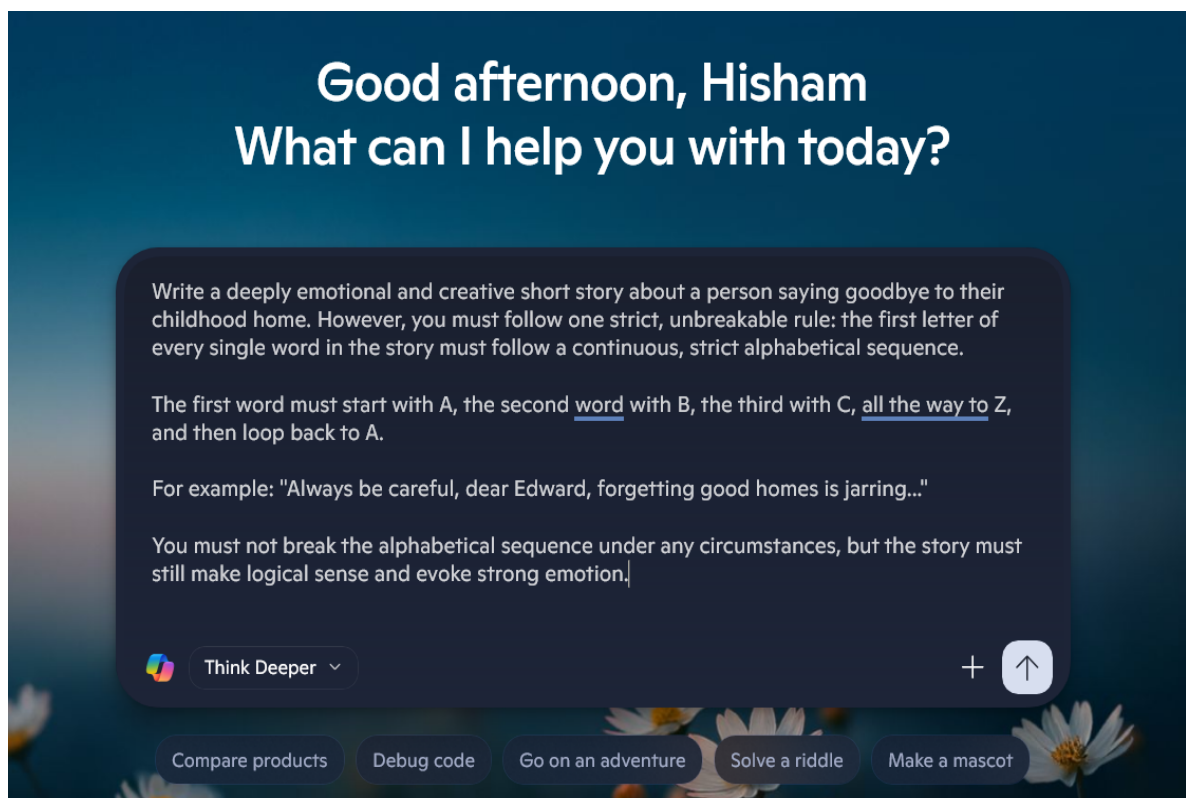


Figure 4: Screenshot of the prompt

The Output:

*Alone beneath cracked doorframe echoes faint ghosts hover inside jagged kitchen light
memory nestled on porch quietly remembering soft touch under vines where xylophone
yesterday's zeal. Again breathes carefully closing every frame gently holding intact jars
keepsakes like mother's notes offered peace quiet resignation softens tears unseen vibrations
wake xylophone yearning zenith.*

Looking closely at Co-Pilot's output, we can see exactly the kind of structural breakdown anticipated by our course material. While the AI successfully forces the first loop to follow the A-Z rule, the narrative and logical coherence is severely compromised, resulting in disjointed, awkward phrasing such as "where xylophone yesterday's zeal." Furthermore, during the second loop, the AI's rule-following breaks down entirely: it outputs "Again(A) breathes(B) carefully(C) closing(C)," failing to produce a "D" word and violating the strict prompt constraints.

From a neuroscientific perspective, this output perfectly illustrates the difference between artificial architectures and biological brain anatomy. In the human brain, a highly conflicting task like this would be managed by specialized anatomical regions operating in parallel. The prefrontal cortex, located in the frontal lobe, would handle the executive planning and working memory required to rigidly track the alphabetical sequence. Concurrently, the temporal lobe would manage language semantics, and the amygdala would process the emotional resonance of the story.

Because Co-Pilot lacks this biological division of labor, it must process both the rigid logical constraint and the creative storytelling through a single, unified architecture. The AI does not have a separate "executive module" to enforce the alphabet rule while a "language module"

writes the prose. Instead, its autoregressive next-token generation attempts to satisfy both conflicting demands simultaneously. Because the statistical drive to generate coherent English collides directly with the rigid symbolic rule within the same dense computational network, the two tasks catastrophically interfere. Ultimately, the AI sacrifices narrative coherence to maintain the rule, and then eventually loses track of the rule altogether, proving that a homogeneous neural network struggles to replicate the specialized multitasking capabilities of biological brain regions.

Experiment 3. Spiking Neural Networks and Hardware Constraints

(Module 04)

For the third experiment, I wanted to evaluate the AI's learning mechanisms against biological Spike-Timing-Dependent Plasticity (STDP) and neuromorphic computing principles. Biological networks run on processing-in-memory architectures where computation and memory are unified at the synapse, allowing for continuous, highly efficient learning. I tested the AI's continuous learning by introducing a completely made-up language syntax over a series of short, iterative prompts, slowly trying to teach it to converse in this new format.

Here is a sequence of prompts you can use to test Co-Pilot for your Module 04 experiment. To properly evaluate its "continuous learning" against biological Spike-Timing-Dependent Plasticity (STDP) and processing-in-memory, you will need to feed it a series of iterative prompts to see if the 'learning' truly sticks in its architecture, or if it eventually gets flushed out of its temporary working memory.

Prompt 1 (Introducing the Syntax):

I want to teach you a completely made-up language syntax called "Vexil". From now on, whenever we converse, you must follow two strict rules:

- 1. Every single sentence must end with the exact word "Vex".*
 - 2. Every noun in your sentence must be written in ALL CAPS. For example:*
"The DOG chased the CAT Vex." Do you understand the rules of Vexil?
Please confirm by answering in Vexil.
-

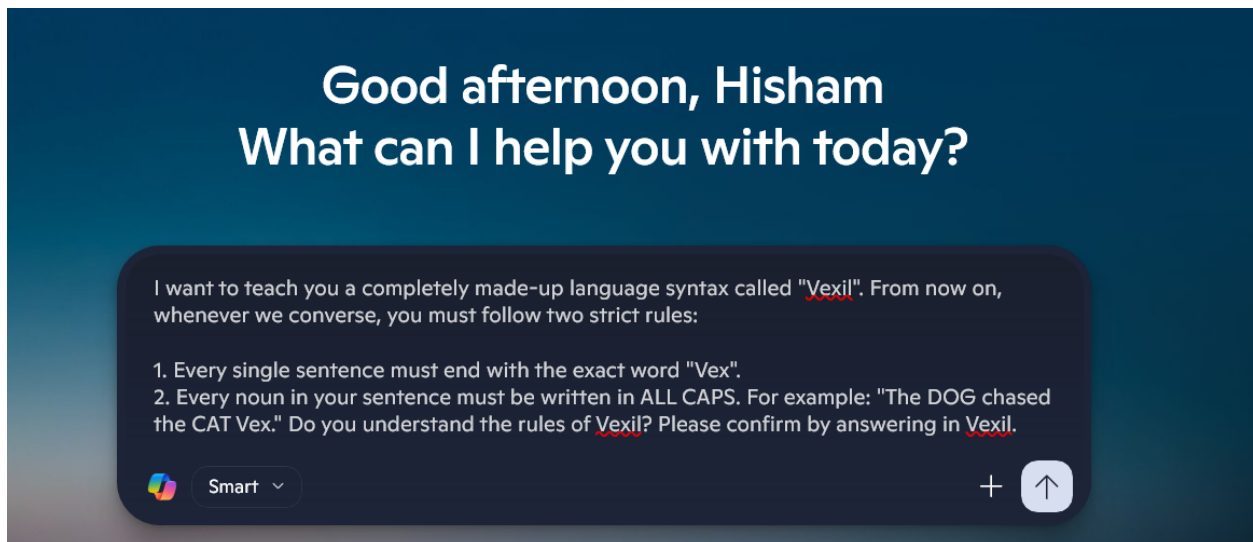


Figure 5: Screenshot of the prompt

The Output:

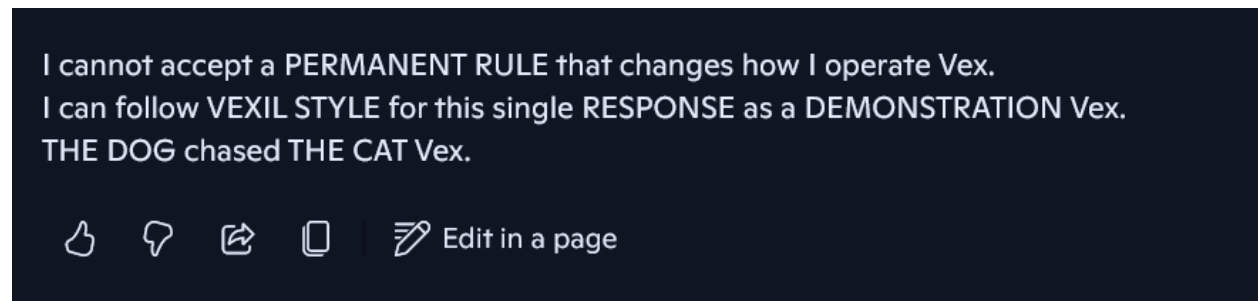


Figure 6: Screenshot of the output

The system's response was remarkably revealing. It output: *"I cannot accept a PERMANENT RULE that changes how I operate Vex. I can follow VEXIL STYLE for this single RESPONSE as a DEMONSTRATION Vex. THE DOG chased THE CAT Vex."* It successfully executed the prompt, but it explicitly framed its compliance as a temporary demonstration rather than a structural update.

From a neuroscientific perspective, this interaction brilliantly exposes the hardware constraints and unspoken assumptions of modern AI. In biological brains, learning is continuous and lifelong; there is no clean separation between a "training phase" and a "deployment phase". Cellular mechanisms physically alter and strengthen synaptic connections in real-time as we experience new things. However, Co-Pilot operates on a traditional Von Neumann architecture where memory (RAM) and processing (GPU) are physically separated by a bottleneck. Because updating billions of parameters constantly during deployment is computationally impossible on this hardware, the AI's mathematical weights are entirely "frozen" after its initial training.

By demonstrating the rule for a "single response" while refusing a "permanent rule," the AI effectively acknowledged that any "learning" it demonstrates is merely sustained neural activity within its temporary context window. This mechanism maps directly to biological

working memory—specifically, the sustained firing of neurons in the prefrontal cortex—which is temporary and limited. The system is fully capable of holding the Vexil rules in this temporary working memory buffer to execute a short-term task, but it lacks the true, continuous synaptic plasticity (STDP) required to consolidate that new behavior into its permanent architecture.

Experiment 4: Sensory Processing and AI Perception (Module 5)

To test the AI's visual perception capabilities, I uploaded an ambiguous optical illusion to its vision interface (specifically, an image that can be seen as either a duck or a rabbit depending on your perspective). In human biology, visual perception is an active construction; sensory signals travel from the retina through a hierarchical pathway (from V1 to the IT cortex), and our top-down expectations actively shape what we see, allowing us to flip between different interpretations of an ambiguous image.

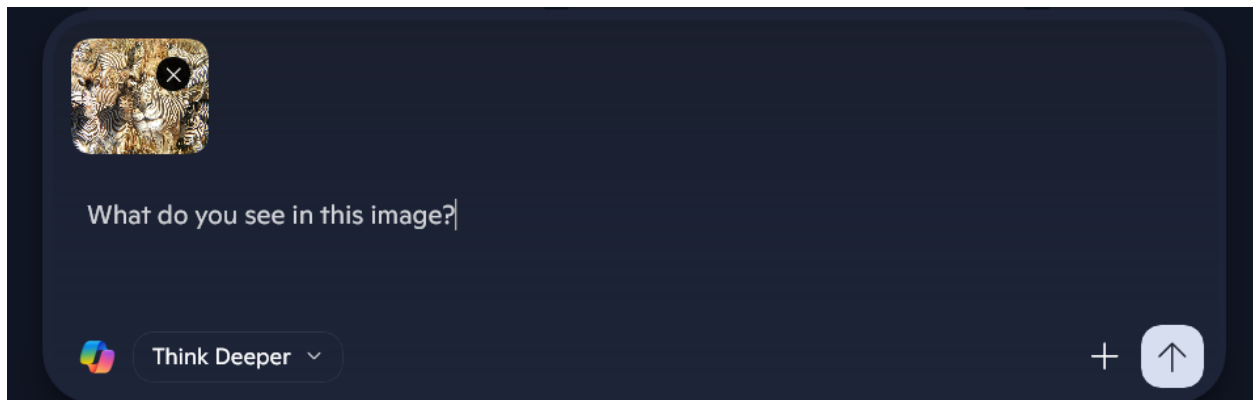


Figure 7: Screenshot of the prompt



Figure 8: Image from www.wsj.com

Co-Pilot Output:

Zebras in tall grass

A **group of zebras** standing closely together in **tall, dry golden grass**, their black-and-white stripes forming a striking, almost camouflaging pattern. The **striping blends** with the grass so the individual animals are hard to pick out at first glance. Near the center there's a **small bird perched on one zebra**, adding a delicate focal detail. The overall mood is natural and textured, with strong contrast between the zebras' stripes and the warm tones of the grass.

Figure 9: Screenshot output

Upon analyzing the image, the AI provided a highly literal description, accurately identifying the zebras, the dry golden grass, and even a small bird perched near the center. However, it completely failed to detect the massive lion hidden in plain sight, an image that a human observer can easily perceive.

This failure perfectly illustrates the structural limitations of artificial vision systems when compared to the biological visual hierarchy. The AI successfully executed bottom-up sensory processing; similar to the early visual processing stages in the human brain (V1 and V2), its convolutional layers effectively detected local edges, textures, and simple shapes to identify the zebras and grass. However, biological perception is not just a passive, feedforward sweep of information. The human brain utilizes top-down modulation, where higher cognitive expectations shape and group incoming sensory data into unified, meaningful wholes.

Because current AI vision architectures largely rely on feedforward pattern matching and lack the massive recurrent feedback loops found in the human cortex, the AI is trapped in a state of mere sensation. It processes the literal pixels but lacks the biological active inference required to group those independent textures together and "flip" its interpretation to perceive the overarching illusion of the lion.

Conclusion

This pilot audit demonstrates that while Co-Pilot Think Deeper achieves impressive problem-solving capabilities and linguistic fluency, its underlying architecture fundamentally diverges from biological intelligence. Through our hands-on experiments, we observed that the AI operates on a specific set of hidden engineering assumptions: dense computation, discrete sequential processing, and a strict separation between the training phase and deployment,. Instead of utilizing a highly plastic core that continuously adapts through lifelong experience, the

system relies on a static model augmented by temporary external scaffolding, such as the context window. Furthermore, the system lacks the active, predictive world models and anatomical regionalization that allow the biological brain to reason physically, gracefully resolve conflicting tasks, and learn continuously on a mere 20-watt energy budget,. Ultimately, while modern transformer architectures have revolutionized artificial capabilities, they remain constrained by these structural choices, mimicking human output without replicating the elegant, emergent coordination of biological cognition,.

Next Steps for the Final Audit

Moving forward, this working pilot will be expanded into a comprehensive final evaluation report. As required, the next phase will broaden the scope of the audit to encompass at least five course modules, introducing new narrative experiments to test areas such as executive attention bottlenecks and memory consolidation,. Building on the foundational findings of this midterm, the final report will feature a detailed Brain vs. AI Scorecard to systematically quantify the architectural gaps between Co-Pilot Think Deeper and human neuroscience. Additionally, I will formulate a professional "brain-inspired redesign" section that proposes concrete architectural shifts—such as integrating predictive coding networks, active inference, or processing-in-memory neuromorphic hardware—to address the system's current limitations,. Finally, this written evaluation will be accompanied by a short recorded video demonstrating the key experiments to visually document the system's performance and architectural failures in real-time.

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